

Switch Adapted Surfing Whale Bubble Machine

MAKER GUIDE

Required Components

1. 	2. 	BOM 1. Play Day Surfing Whale Bubble Machine 2. 3.5 mm Mono Jack and Nut 3. 22 AWG Wire 4. 4 AA Batteries Refer to the Tool and Component List for purchasing details.
3. 	4. 	

Required Tools

- Phillips Screwdriver
- Soldering iron
- Wire stripper
- Flush cutter
- Drill and ¼" drill bit
- Sharpie (optional)

Required Personal Protective Equipment (PPE)

- Safety Glasses

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Assembly Instructions

1. Carefully remove the toy from the packaging. The toy will be put back in after it is adapted.

Using a screwdriver, turn the 5 screws indicated in the red circles to open the toy. Set the screws aside in a safe place.

Note: The toy includes 23 screws in total. All screws are identical except for the one used to secure the tail.



2. Remove the tail (inner battery holder) by pressing and holding the red button on the underside of the whale.

Remove the front and back bathing suit pieces.



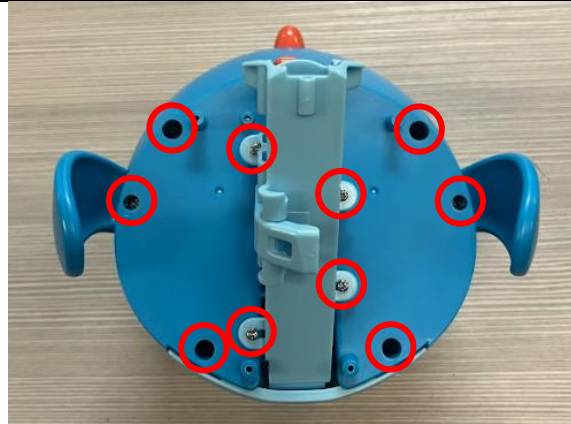
3. Unscrew the 4 screws in red circle and set aside.



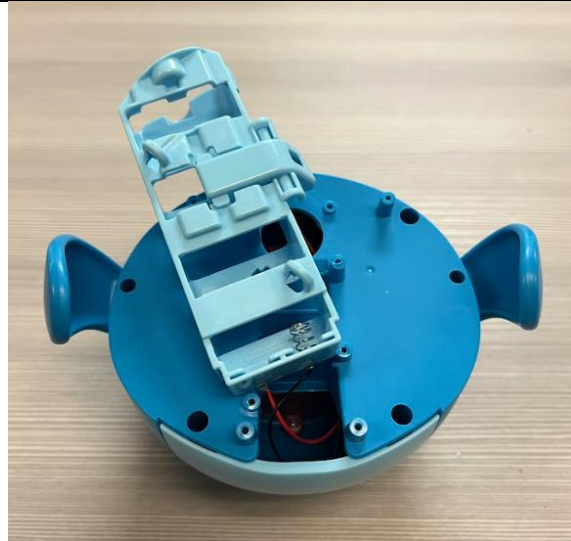
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4. Unscrew the 10 screws in the red circles and set aside.



5. Gently take out the battery cartridge. Do not tug or else the wires may accidentally disconnect.



6. Remove the belly by sliding it up. Set it aside.



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7. Take off the bubble wand by gently tugging it towards you.



8. Pry open the two halves of the whale by starting from one open end and working your way around.



9. Unscrew the 2 screws securing the motor and set it aside.



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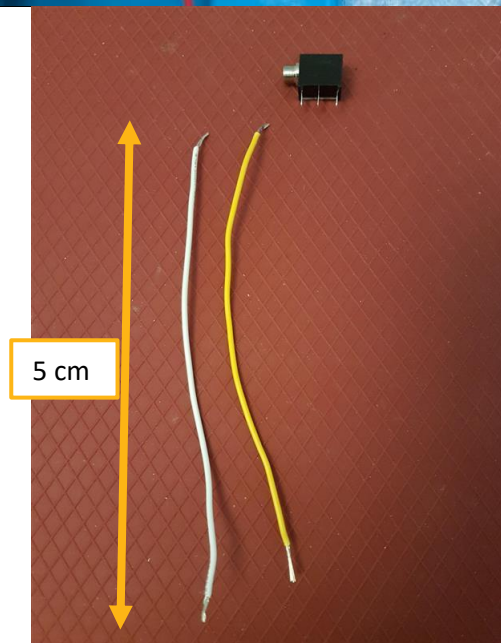
10. Make sure the switch is in the OFF position. This will be helpful later.

Unscrew the 2 screws securing the slide switch and set it aside.



11. Next, we will prepare the mono jack and wires.

Cut 2 pieces of wire each 5 cm in length.



12. Use wire strippers to strip approximately 0.5 cm off each end of the wires.



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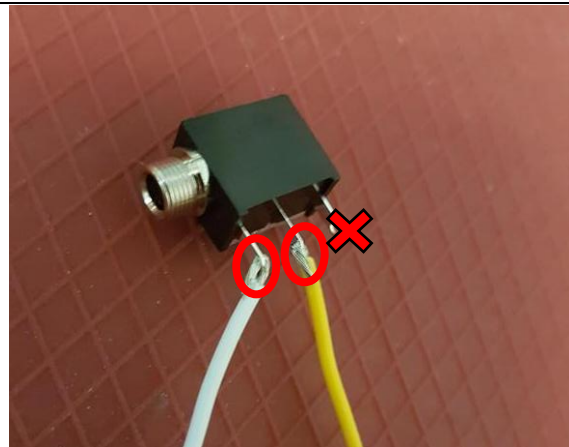
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13. Pick one wire, slide exposed wire into first metal arm on mono jack (closest to the jack end).

With second wire, slide one end of exposed wire into the middle metal arm on mono jack.

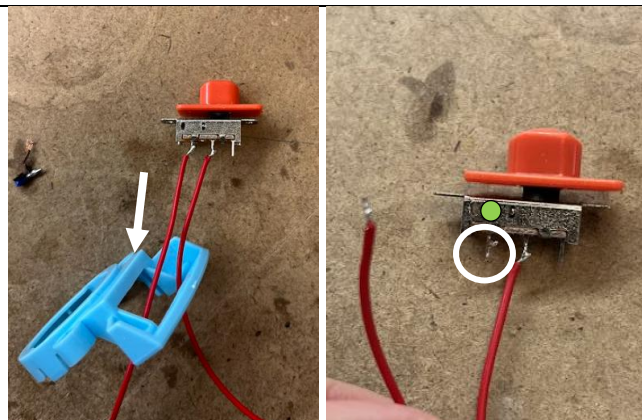
Solder the wire to the mono jack.

Note: Check that you have the correct metal arms of mono jack. Refer to the picture.



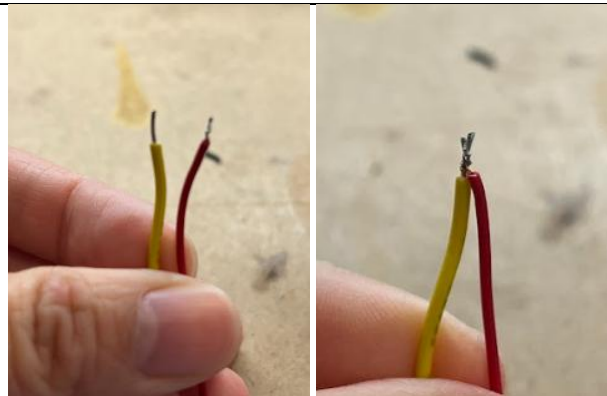
14. Take out the slide switch from the plastic holder and slide it out of the way.

Detach the left wire by placing the tip of the soldering iron to the solder point. It's recommended to mark the left lead with a Sharpie as you will reattach the wire to this point later.



15. Strip the detached wire by 0.5cm to expose more wire.

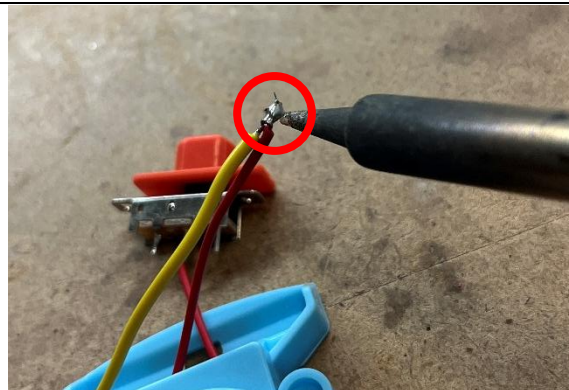
Twist together the detached wire from Step 14 and 1 wire from the mono jack assembly from Step 13 (any wire).



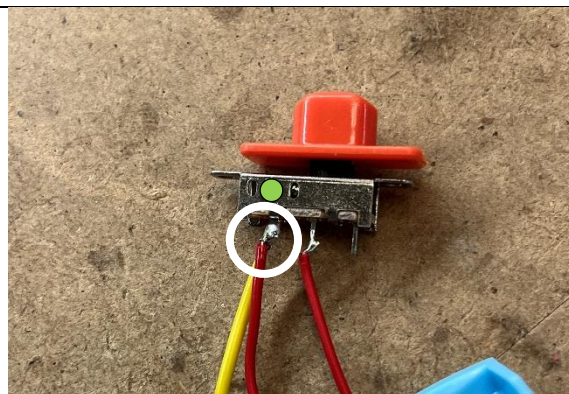
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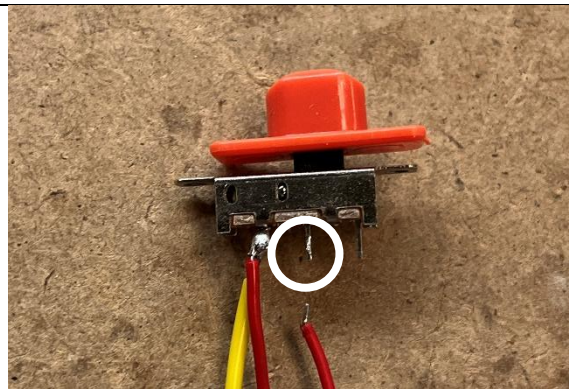
16. Tin the wires by adding solder to the ends.



17. Solder the wires to the left slide switch lead (previously marked with the Sharpie).



18. Detach the middle wire by placing the tip of the soldering iron to the solder point.

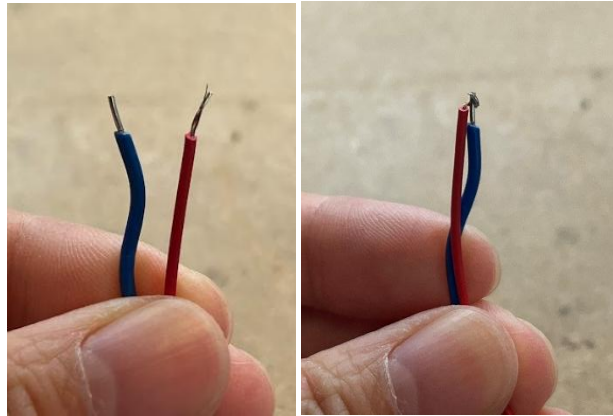


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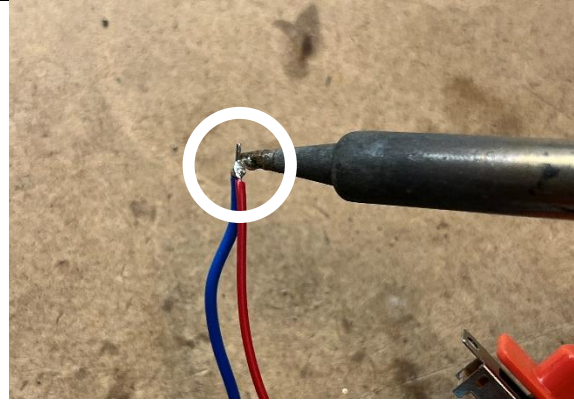
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19. Strip the detached wire by 0.5cm to expose more wire.

Twist together the detached wire from Step 18 and the remaining wire from the mono jack assembly from Step 13.



20. Tin the wires by adding solder to the ends.



21. Solder the wires to the middle slide switch lead.



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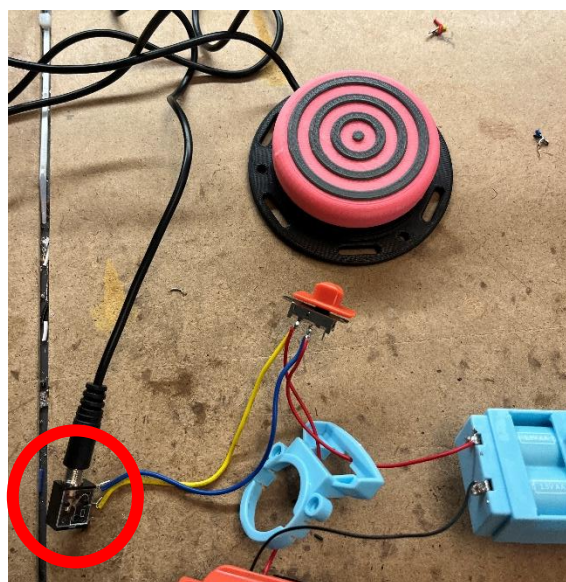
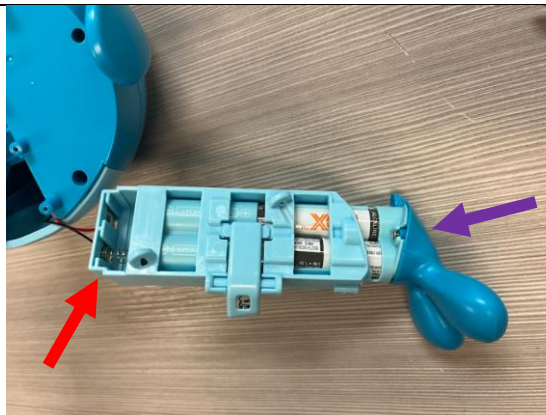
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22. Test if the assembly works.

First, insert 4 AA batteries into the inner battery holder (top image, purple arrow) and slide it into the outsider battery compartment (red arrow). It should click into place.

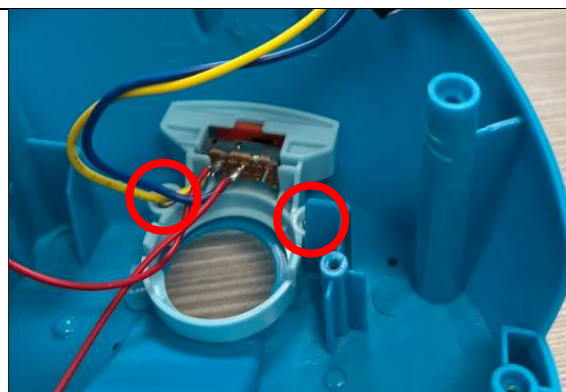
Then, plug an assistive switch into the mono jack (red circle). When the assistive switch is pressed, the motor should spin. This indicates that it was soldered correctly. The motor should also automatically spin in the ON position.

If the motor starts spinning automatically in both the ON and OFF positions (when the assistive switch is not pressed), there may be a soldering issue. Check that all solder joints were done cleanly and in the right position, and correct as needed.



23. Insert the slide switch back in the light blue plastic holder. Ensure the switch orientation matches the ON / OFF label on the toy.

Secure with 2 screws.



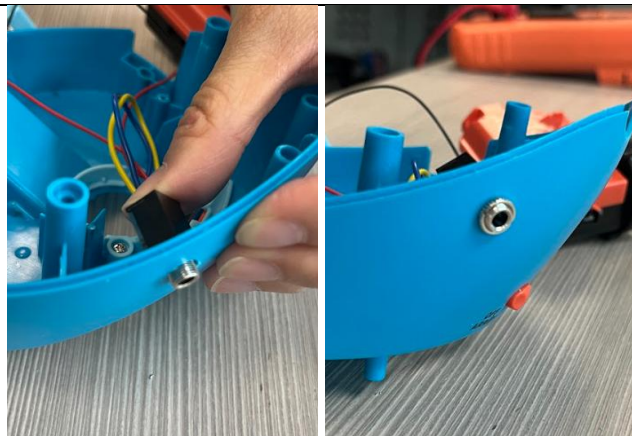
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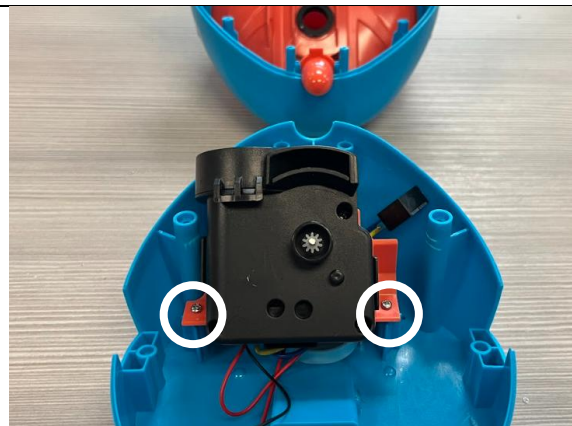
24. With ¼ inch drill bit, drill a hole in the toy, shown in the red circle.



25. Insert the mono jack through the hole and secure in place by twisting the metal ring onto the jack from the outside of the toy.



26. Secure the motor back in place with 2 screws.



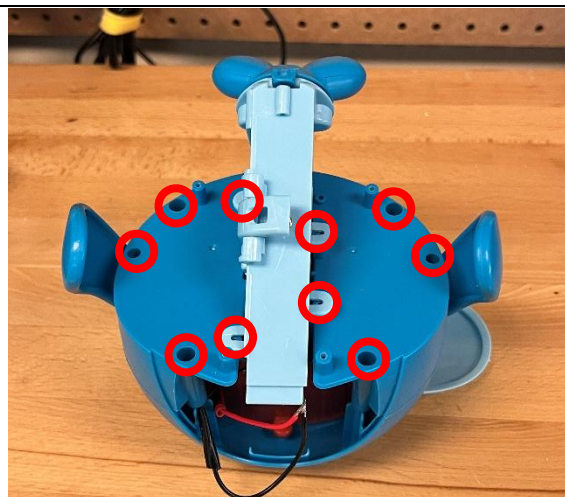
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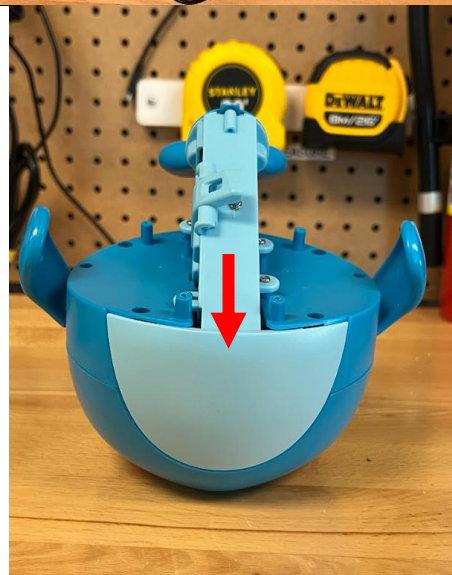
27. Reassemble the two halves of the toy.



28. Screw the 10 screws back in place (4 for the battery holder, 6 for the whale body).



29. Slide the belly back in place.

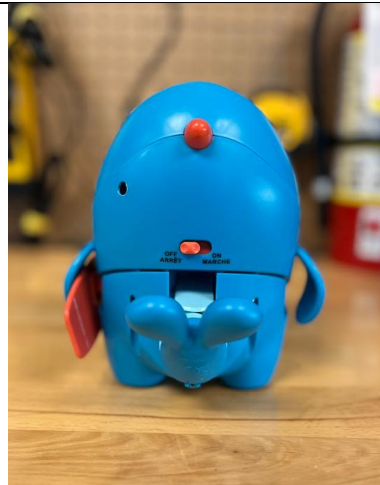


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30. Combine the top and bottom halves of the whale. There will be a small gap between the two pieces – this is normal.

Use 4 screws to secure the two halves together.



31. Take out the inner battery holder (tail piece) by pressing and holding the red button on the underside of the whale.



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32. Fit both pieces of the bathing suit on the whale.



33. Insert the inner battery holder (tail piece) back. Secure everything together with 5 screws.

Note that the screw that secures the tail piece is different than the other screws.



34. Insert the bubble wand back in place.



35. You can check the toy again with an assistive switch to make sure it works once it is reassembled. If it does not work, check the connections inside. This includes looking at where wires may have been pinched when the toy was reassembled.

